INCH-POUND

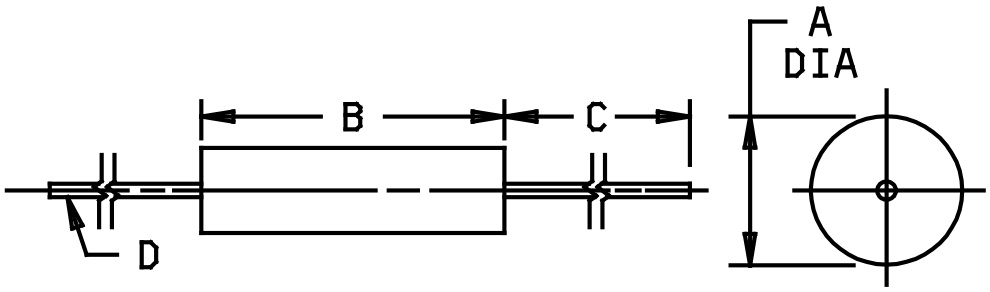
MIL-PRF-39010/10C
W/AMENDMENT 2
14 December 2022
SUPERSEDING
MIL-PRF-39010/10C
W/AMENDMENT 1
6 September 2018

PERFORMANCE SPECIFICATION SHEET

# COILS, FIXED, RADIO FREQUENCY, MOLDED, MICROMINIATURE (FERRITE CORE), ESTABLISHED RELIABILITY AND NONESTABLISHED RELIABILITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-39010.




| Letter | Dimensions are in inches with metric equivalents (mm) in parentheses | Dimensions are in inches with metric equivalents (mm) in parentheses |
|---|---|---|
| | Minimum | Maximum |
| A | .085 (2.16) | .105 (2.67) |
| B | .240 (6.10) | .260 (6.60) |
| C | 1.250 (31.75) | 1.626 (41.30) |
| D | .0185 (0.470) | .0215 (0.546) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.

FIGURE 1. Dimensions and configuration.

AMSC N/A FSC 5950

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Material: Ferrite core.

Weight: 0.01 ounce maximum.

Operating temperature range: -55°C to +105°C.

Dielectric withstanding voltage: MIL-STD-202-301; test voltage of 1,000 V rms.

Barometric pressure: MIL-STD-202-105, test condition C, 70,000 feet with a test voltage of 200 V rms.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Inductance tolerance: See table I.

Q values: See table I.

Self-resonant frequency: See table I.

DC resistance: See table I.

Temperature rise: 15°C.

Terminal pull: 5 pounds.

Part or Identifying Number (PIN): M39010/10 (dash number from table I).

Supersession data: This specification supersedes MS75085.



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TABLE I. Electrical characteristics (initial) and dash numbers.

| Dash number 1/ | Inductance μH | Inductance tolerance ± percent | Q minimum | Test frequency (MHz) | Self-resonant frequency minimum (MHz) | DC resistance (25°C) maximum (ohms) | Rated dc current (mA) 2/ |
|---|---|---|---|---|---|---|---|
| A300** | 30.0 | 5 | 45 | 2.5 | 24 | 3.4 | 130 |
| A330** | 33.0 | 5, 10 | 45 | 2.5 | 24 | 3.4 | 130 |
| A360** | 36.0 | 5 | 45 | 2.5 | 22 | 3.6 | 125 |
| A390** | 39.0 | 5, 10 | 45 | 2.5 | 22 | 3.6 | 125 |
| A430** | 43.0 | 5 | 45 | 2.5 | 20 | 4.5 | 110 |
| A470** | 47.0 | 5, 10 | 45 | 2.5 | 20 | 4.5 | 110 |
| A510** | 51.0 | 5 | 45 | 2.5 | 18 | 5.7 | 100 |
| A560** | 56.0 | 5, 10 | 45 | 2.5 | 18 | 5.7 | 100 |
| A620** | 62.0 | 5 | 45 | 2.5 | 15 | 6.7 | 92 |
| A680** | 68.0 | 5, 10 | 50 | 2.5 | 15 | 6.7 | 92 |
| A750** | 75.0 | 5 | 50 | 2.5 | 14 | 7.3 | 88 |
| A820** | 82.0 | 5, 10 | 50 | 2.5 | 14 | 7.3 | 88 |
| A910** | 91.0 | 5 | 50 | 2.5 | 13 | 8.0 | 84 |
| A101** | 100.0 | 5, 10 | 50 | 2.5 | 13 | 8.0 | 84 |
| A111** | 110.0 | 5 | 30 | .79 | 12 | 13.0 | 66 |
| A121** | 120.0 | 5, 10 | 30 | .79 | 12 | 13.0 | 66 |
| A131** | 130.0 | 5 | 30 | .79 | 11 | 15.0 | 61 |
| A151** | 150.0 | 5, 10 | 30 | .79 | 11 | 15.0 | 61 |
| A161** | 160.0 | 5 | 30 | .79 | 10 | 17.0 | 57 |
| A181** | 180.0 | 5, 10 | 30 | .79 | 10 | 17.0 | 57 |
| A201** | 200.0 | 5 | 30 | .79 | 9 | 21.0 | 52 |
| A221** | 220.0 | 5, 10 | 30 | .79 | 9 | 21.0 | 52 |
| A241** | 240.0 | 5 | 30 | .79 | 8 | 25.0 | 47 |
| A271** | 270.0 | 5, 10 | 30 | .79 | 8 | 25.0 | 47 |
| A301** | 300.0 | 5 | 30 | .79 | 7 | 28.0 | 45 |
| A331** | 330.0 | 5, 10 | 30 | .79 | 7 | 28.0 | 45 |
| A361** | 360.0 | 5 | 30 | .79 | 6.5 | 35.0 | 40 |
| A391** | 390.0 | 5, 10 | 30 | .79 | 6.5 | 35.0 | 40 |

See footnotes at end of table.



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TABLE I. Electrical characteristics (initial) and dash numbers- continued.

| Dash number 1/ | Inductance μH | Inductance tolerance ± percent | Q minimum | Test frequency (MHz) | Self-resonant frequency minimum (MHz) | DC resistance (25°C) maximum (ohms) | Rated dc current (mA) 2/ |
|---|---|---|---|---|---|---|---|
| A431** | 430.0 | 5 | 30 | .79 | 6 | 42.0 | 36 |
| A471** | 470.0 | 5, 10 | 30 | .79 | 6 | 42.0 | 36 |
| A511** | 510.0 | 5 | 30 | .79 | 5 | 46.0 | 35 |
| A561** | 560.0 | 5, 10 | 30 | .79 | 5 | 46.0 | 35 |
| A621** | 620.0 | 5 | 30 | .79 | 4 | 60.0 | 30 |
| A681** | 680.0 | 5, 10 | 30 | .79 | 4 | 60.0 | 30 |
| A751** | 750.0 | 5 | 30 | .79 | 3.8 | 65.0 | 29 |
| A821** | 820.0 | 5, 10 | 30 | .79 | 3.8 | 65.0 | 29 |
| A911** | 910.0 | 5 | 30 | .79 | 3.4 | 72.0 | 28 |
| A102** | 1,000.0 | 5, 10 | 30 | .79 | 3.4 | 72.0 | 28 |

1/ The complete dash number will include two additional letters (indicated by **). The first additional letter will indicate the inductance tolerance and the second additional letter will indicate the product level (e.g., C, M, P, R, S) and will be added to the end of the dash number.
2/ The rated dc current is based on 90°C ambient temperature with a 15°C rise.



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TABLE II. Electrical characteristics (final). 1/

| Inspection group | Allowable variation from initial measurement | Allowable variation from initial measurement | Allowable percent from specified minimum value in electrical characteristics (initial) table | Allowable percent from specified minimum value in electrical characteristics (initial) table |
|---|---|---|---|---|
| | Inductance (percent) | DC resistance | Self-resonant frequency | Q |
| Qualification inspection | | | | |
| Group II | ±2 | | | -10 |
| Group IV | ±10 | (±5% +.001 ohm) | 2/ | -20 |
| Group V | ±5 | (±2% +.001 ohm) | 2/ | -15 |
| Quality conformance inspection Group B | | | | |
| Subgroup 1 | ±5 | ±(2% +.001 ohm) | 2/ | -15 |
| Subgroup 3 | ±2 | | | -10 |
| Subgroup 4 | ±10 | ±(5% +.001 ohm) | 2/ | -20 |

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±( _ percent +.01 μH).
2/ The self-resonant frequency shall be not less than the value specified in table I.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-39010, this document references the following:

MIL-STD-202-105, MIL-STD-202-301



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Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS, SH
Air Force - 19

(Project 5950-2022-078)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.



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